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1.0 INTRODUCTION AND CASE STUDY DEFINITION

1.1 Description of the selected scientific or industrial problem

Student academic performance is a critical indicator of the effectiveness of higher education systems and directly influences graduate employability, institutional reputation, and national human capital development. Universities increasingly rely on data-driven decision-making to identify factors that affect student success and to design targeted academic interventions.

This study presents a case study based on a survey dataset collected from students of Universiti Malaysia Kelantan (UMK). The dataset contains demographic, socio-economic, academic, behavioral, and lifestyle variables such as gender, age, faculty, study habits, attendance, GPA/CGPA, use of technology, co-curricular involvement, and health-related factors.

1.2 Relevance of the problem to industrial practices

Main Problem:

The lack of data-driven understanding of how demographic, academic, and behavioral factors influence student academic performance among Universiti Malaysia Kelantan students limits the effectiveness of early intervention strategies and academic decision-making.

To analyze and visualize the factors influencing student academic performance among Universiti Malaysia Kelantan students.

1.2.1 Demographic

- Academic performance is shaped by both demographic and academic factors. At Universiti Malaysia Kelantan (UMK), students differ in gender, age, race, relationship status, and academic behaviors such as attendance and CGPA/GPA. This study aims to explore these relationships to identify key predictors of academic success and inform strategies to improve student outcomes.
- To examine the influence of demographic factors on students academic performance at Universiti Malaysia Kelantan.

1.2.2 Study and Lifestyle

- Study habits, social media usage and health factors are aspects of lifestyle that could have influenced the academic performance of UMK students. However, the relationship between these factors and students' academics is still poorly explored through visual analysis. Therefore, this study aims to visualize and

understand the relationship between lifestyle factors and understand the relationship between lifestyle factors and students' academic performance.

- To analyse the relationship between study habits, social media usage and health factors with the academic performance of UMK students.

1.2.3 Learning Mode

- Learning mode preference plays an important role in shaping students' learning experiences and academic outcomes at UMK. However, the relationship between learning mode preference, year of study, and academic performance remains insufficiently explored through visual analysis, limiting clear understanding of performance patterns and potential academic risks. Therefore, this study aims to visualize and examine how learning mode preference and year of study relate to academic performance among UMK students.
- To analyze the relationship between learning mode preference, year of study, and academic performance of UMK students.

1.2.4 Skills and Activities

- While UMK emphasizes holistic student development, there is a lack of empirical evidence regarding how the time invested in non-academic skill development and participation in co-curricular activities correlates with academic success. Students often face a dilemma in balancing extracurricular commitments with their studies, leading to uncertainty about whether these activities enhance academic performance through soft skill acquisition or hinder it due to time constraints
- To analyze how skill development levels and co-curricular participation influence students' academic performance of UMK students using visual analytics.

1.3 Justification for the use of scientific visualization

The use of scientific visualization is justified due to the multidimensional and heterogeneous nature of the survey dataset, which contains both numerical and categorical variables. Scientific visualization enables the transformation of complex data into intuitive graphical representations, allowing patterns, trends, and anomalies to be identified more effectively than through tabular data alone. Visualization techniques facilitate comparisons across student groups, such as faculty, study habits, or socio-economic background, and help reveal relationships between behavioral factors and academic performance. As a result, visualization serves as a powerful tool for supporting decision-making processes and communicating findings to both technical and non-technical stakeholders.

2.0 ETHICAL CONSIDERATIONS AND DATA COLLECTION

2.1 Survey or data collection methodology

This study used a structured online questionnaire and a quantitative survey-based methodology to look into the variables affecting UMK students' academic performance. The survey approach was chosen because it is effective in gathering standardized data that can be analyzed and visualized.

The questionnaire was divided into five sections: demographics, academic information, learning behavior and study habits, skills and development activities and also health and personal well-being. The background, academic history, study habits, skill development, and health state of the students were all gathered in these parts. To ensure uniformity and convenience of analysis, the majority of the questions were closed-ended, with some numerical answers. The online model decreased manual data entry errors and made it simple for responders to finish the survey.

2.2 Target respondents, sampling approach, and sample size

The target respondents of this study were undergraduate students from all faculties of Universiti Malaysia Kelantan (UMK). This inclusive approach ensured representation from several types of academic backgrounds. A non-probability convenience sampling method was used. The questionnaire link was shared in WhatsApp group chats, personal messages, and status updates, allowing for voluntary participation. Data gathering lasted three weeks and received 103 valid responses. This sample size was considered sufficient for identifying broad patterns connected to student academic performance.

2.3 Ethical considerations

Ethical principles were strictly observed throughout the study. Participation was voluntary, and respondents were informed of the study's purpose through an introductory statement at the beginning of the questionnaire. All responses were kept private, and no personally identifiable information was collected. The data was only used for academic purposes relating to this course. Respondents were encouraged to provide honest answers and were informed that they could withdraw from the survey at any time without any consequences. Contact details of the researchers were provided to ensure transparency and address any enquiries.

3.0 DATA CLEANING AND PREPROCESSING

3.1 Identification and treatment of missing, inconsistent, or noisy data

Initial data screening was conducted using summary functions, unique value checks and duplicate record detection to identify missing, inconsistent or redundant data issues. Irrelevant or unused variables in the visualization and study such as timestamp, scholarship, repeat courses, study session and several others were removed. In addition, non-standard data formats such as Year of Study values in text form were standardized to numeric format to ensure data consistency.

3.2 Data transformation, normalization, or categorization

Several data transformations were implemented to facilitate analysis and visualization. These included changing column names to a more consistent format, such as Year Of Study to Year_of_Study. In addition, the calculation of midpoint values for range-based variables such as GPA, CGPA, attendance, age and family income. A coding process was also carried out for categorical variables such as health issues (No = 0, Yes = 1) without removing the original values. In addition, a categorization process was carried out on several lifestyle variables to facilitate interpretation. Study hours, social media usage and skill development hours were categorized into low, medium and high levels based on the daily time range. Family income was categorized into B40, M40 and T20 groups based on the specified income range, while the income midpoint value was also calculated. Standardization of faculty names was also carried out by shortening faculty names to official abbreviations. All these steps helped ensure that the data was more structured, easy to visualize and aligned with the study analysis objectives.

For the Skills variable, the data cleaning and classification process was carried out using a rule-based approach. The skills text data was first normalized by converting all letters to lowercase and removing symbols and special characters to ensure text uniformity. Next, the skills were categorized based on matching keywords to several defined skill domains. Skills related to communication, leadership, teamwork and interpersonal abilities were classified as Soft Skills, while skills involving programming, software development, data analysis, networking, cybersecurity and hardware support were categorized as technical skills according to their respective domains. Skills that were general, non-specific or did not match any set of keywords were classified under the Others category. This approach was chosen because the skills data is in the form of free text and requires a classification method that is consistent, transparent and suitable for data analysis and visualization.

4.0 SCIENTIFIC VISUALIZATION TECHNIQUES

4.1 Demographic (Hidayah)

- A **violin plot** is used to visualize the distribution of CGPA midpoint by gender. Gender is mapped to the x-axis and CGPA midpoint to the y-axis. The width of each violin represents the density of the data, while the internal markers indicate the median and interquartile range, allowing comparison of central tendency, spread, and distribution shape between genders.
- A **histogram** is applied to examine GPA midpoint distributions across relationship status and gender, which is appropriate for continuous numerical data; GPA midpoint is shown on the x-axis, frequency on the y-axis, with colour or faceting to distinguish gender and a dropdown to select relationship status for clearer comparison.
- A **line chart** is used to analyse CGPA trends across different years of study, as it effectively shows progression over an ordered variable; CGPA midpoint and GPA midpoint are mapped to the axes, with separate lines representing year-of-study midpoints to compare academic trends over time.
- A **bar chart** compares the average CGPA midpoint across income categories (B40, M40, and T20), with income groups on the x-axis and average CGPA midpoint on the y-axis, where bar height directly encodes performance differences between socioeconomic groups.
- A **heatmap** explores the relationship between study hours per day, attendance percentage, and CGPA midpoint by living arrangement, plotting study hours and attendance on the axes and encoding CGPA midpoint using colour intensity with a sequential gradient, while living arrangement is incorporated through grouping or dropdown-based faceting.
- Across all visualizations, perceptual effectiveness is ensured through the use of distinct but muted colours for categorical variables, consistent numerical scales to avoid distortion, even spacing of categories, simple and uncluttered layouts with clear labels, legends, and titles, minimal unnecessary gridlines, and intuitive visual encodings that enable quick and accurate pattern recognition.

4.2 Study and Lifestyle (Syazana)

- **A bar chart** is used to illustrate average GPA across study time categories, as it allows for clear comparisons between different levels of study time and their relationship to academic performance. The x-axis represents study time categories, while the y-axis represents average GPA. The height of each bar indicates the difference in academic performance across study time groups.
- **A box plot** is used to analyze the distribution of academic performance based on categories of daily social media use. This graph is suitable because it displays the median, variation, and extreme values of GPA, thus helping to understand the differences and imbalances in academic performance among students. The x-axis represents the category of daily social media use, while the y-axis represents the midpoint value of GPA. The box indicates the interquartile range, the center line represents the median, and the extreme values reflect the variation in academic performance.
- **A bar graph** is used to compare the average academic performance between students with health issues and students without health issues, and to assess the influence of health factors on academic achievement. The x-axis represents the status of the students' health issues, while the y-axis represents the average academic performance. The height of the bar indicates the difference in academic performance between the two groups.
- **A line chart** is used to visualize the trend in the relationship between attendance rates and student academic performance. This graph is appropriate because it highlights the changes and patterns of GPA performance across different attendance levels. The x-axis represents attendance rates (midpoint), while the y-axis represents the average GPA. The lines connecting the data points show the trend in academic performance changes by attendance level.
- **A correlation heatmap** is used to display the relationships between several study variables simultaneously. This visualization is suitable because it facilitates comparison of the strength and direction of the relationship between daily study hours, daily social media use, attendance, and GPA through color intensity. The x-axis and y-axis represent the study variables in the form of midpoints, while colors are used to indicate the strength and direction of the correlation between the variables.
- Colors are used consistently to avoid confusion between graphs, while axis scales are clearly defined to ensure fair data comparisons. The layout of one graph per view improves visual clarity and reader focus. The use of line graphs helps to highlight trends in data changes, while bar graphs facilitate comparisons between categories. In addition, heatmaps use differences in color intensity to facilitate interpretation of the strength of relationships between variables.

4.3 Learning Mode (Fatin Athirah)

- **A pie Chart** is used to illustrate the overall distribution of learning mode preferences (online, offline, and hybrid) among UMK students, as it effectively represents proportions within a single categorical variable; learning mode categories are encoded as segments of the pie, the size of each segment reflects the proportion of students selecting each mode, and colour is used to clearly distinguish between the different learning modes.
- **A bar Chart** is employed to compare the average GPA midpoint across different learning modes, which is suitable for comparing mean values between distinct groups; learning mode is mapped to the x-axis, average GPA midpoint is mapped to the y-axis, and the height of each bar directly encodes academic performance differences, allowing for straightforward comparison of short-term academic outcomes across teaching approaches.
- **A box Plot** is applied to analyse the distribution and variability of CGPA midpoint across learning modes, as it provides a clear summary of central tendency and dispersion; learning mode is shown on the x-axis, CGPA midpoint on the y-axis, the box represents the interquartile range, the median is indicated by a central line, and whiskers encode the overall spread of values, enabling comparison of performance stability between learning environments.
- **A stacked Bar Chart** is used to examine learning mode preferences across different years of study, which is effective for analysing changes in categorical distributions over time; year of study is mapped to the x-axis, learning mode is represented as stacked colour segments within each bar, and the height of each segment corresponds to the number of students preferring a specific learning mode, allowing both overall and within-year comparisons.
- **A heatmap** is utilised to explore the relationship between learning mode, year of study, and average CGPA midpoint, as it effectively visualises multivariate patterns; year of study is mapped to the x-axis, learning mode to the y-axis, and average CGPA midpoint is encoded using a sequential colour gradient, where higher values are represented by more intense shades to highlight high- and low-performing combinations.
- Across all visualisations, perceptual effectiveness is ensured through the use of distinct and consistent colours for learning modes, sequential colour gradients for numerical values, consistent axis scales to support fair comparisons, uncluttered layouts with each visualisation presented separately, and clear titles, labels, and legends that enhance readability and support quick and accurate interpretation of patterns.

4.4 Skills and Activities (Syazwanie)

- **A KDE (Kernel Density Estimation) plot** is used to compare the distribution of CGPA between active and non-active students, as it provides a smooth estimation of the probability density function to reveal patterns that histograms might obscure; CGPA is mapped to the x-axis, density to the y-axis, and student activity status is distinguished by color-coded lines, allowing for a clear comparison of where academic performance peaks and how the spread differs between the two groups.
- **A grouped bar chart** compares the average CGPA midpoint across skill development categories and co-curricular involvement, which is effective for showing the interaction between two categorical variables; skill development level is mapped to the x-axis, CGPA midpoint to the y-axis, and bar color identifies co-curricular status (Yes/No), with side-by-side placement directly encoding the performance difference within each skill tier.
- **A stacked percentage bar chart** is applied to visualize the proportion of different CGPA ranges within various skill categories, ensuring a fair comparison across categories regardless of the total student count; skill category is mapped to the y-axis, percentage (0-100%) to the x-axis, and CGPA range segments are color-coded within each bar to show the internal composition and distribution of academic success across disciplines.
- **A multi-series line chart** is used to track CGPA progression over the four years of study across different skill levels, as it effectively highlights trends and the rate of change over an ordered temporal variable; year of study is mapped to the x-axis, CGPA midpoint to the y-axis, and separate lines distinguished by both color and markers represent skill development categories to illustrate how academic performance evolves throughout the degree.
- **A heatmap** explores the relationship between skill development hours and co-curricular participation on average CGPA, providing an intuitive matrix-based summary of performance; skill development category is plotted on the y-axis, co-curricular status on the x-axis, and average CGPA is encoded through color intensity using a sequential blue gradient, where darker shades represent higher performance and text labels provide precise numerical data for quick reference.
- Across all visualizations, perceptual effectiveness is ensured through the use of distinct but professional color palettes (such as the sequential blue for the heatmap and categorical hues for student status), consistent numerical scales to prevent data distortion, and even spacing of categories. The layouts are kept simple and uncluttered with clear axis labels, legends, and titles, while the use of direct labeling (as seen in the heatmap and bar charts) minimizes the cognitive load required to interpret specific data points. This combination of visual encodings enables quick, accurate pattern recognition and effectively communicates the correlations between student habits and academic outcome.

5.0 INTERPRETATION AND ANALYSIS

5.1 Demographics (Hidayah)

- **The violin plot** shows that female students generally achieve slightly higher and more consistent CGPA values, while male students exhibit a wider spread of scores but no extreme outliers, suggesting a moderate influence of gender on CGPA.
- **The histogram** illustrates GPA distributions across relationship statuses, where single students show the widest GPA spread and married students have a more concentrated distribution at higher GPA levels; however, strong overlap between genders and the absence of clear anomalies indicate that relationship status is not a dominant factor affecting GPA.
- **The line chart** reveals a positive correlation between GPA and CGPA across all years of study, with CGPA increasing as students progress and third-year students achieving the highest values, even outperforming fourth-year students, possibly due to final-year academic pressure.
- **The bar chart** shows that M40 students record the highest average CGPA, followed closely by B40 students, while T20 students have the lowest average CGPA, highlighting that higher income does not necessarily result in better academic performance and that resilience and motivation play important roles.
- **The heatmap** demonstrates that higher attendance consistently leads to higher CGPA across most living arrangements, with students living in hostels and with friends showing clearer performance patterns; hostel students display both high and low CGPA values, indicating high variability, while patterns for students living alone are less clear due to limited data.
- Overall, these findings connect to the problem context by showing that academic performance is shaped by a combination of gender, academic experience, study habits, attendance, and living environment rather than socioeconomic factors alone.

5.2 Study and Lifestyle (Syazana)

- **The bar graph** shows an increase in average GPA in line with the increase in study hours, reflecting a positive trend between study habits and academic performance. This finding supports the case study problem that study habits play an important role in influencing the academic achievement of UMK students.
- **The box plot graph** shows the variation in academic performance across levels of social media use, although the median GPA is almost the same for each category. The greater variation in the high social media use group reflects the inconsistency of academic performance. This finding reinforces the case study problem regarding the influence of digital lifestyle on the stability of students' academic achievement.
- **The bar chart** shows that students without health issues recorded higher average GPAs than students with health issues, showing a consistent pattern of differences in the data. This finding suggests that health factors are part of a lifestyle that has the potential to influence students' academic performance.
- **The line graph** shows that GPA tends to increase as attendance increases, although there is a slight drop around 70% attendance. This pattern suggests that consistent student engagement has a positive impact on academic performance, emphasizing the importance of engagement in the learning process.
- **The correlation heatmap** shows the relationship between study factors, with study time and attendance having a stronger relationship with GPA than social media use. A positive relationship exists between study time, attendance and GPA, while social media use shows a weaker relationship. This analysis supports the study context that academic performance is influenced by a combination of several lifestyle factors and study habits.

5.3 Learning Mode (Fatin Athirah)

- **The pie chart** indicates that Hybrid learning is the most preferred learning mode among UMK students, followed by Offline learning, while Online learning is the least favoured. This pattern suggests that students value the flexibility of hybrid learning while still appreciating opportunities for face-to-face interaction, highlighting the importance of balanced learning environments in supporting student engagement.
- **The bar chart** shows that average GPA values across different learning modes are relatively similar, with Offline learning recording a slightly higher average GPA compared to Hybrid and Online modes. However, the differences are minimal, indicating that learning mode alone does not strongly influence short-term academic performance and that students are generally able to adapt to various instructional approaches.
- **The box plot** reveals differences in CGPA variability across learning modes, where Offline learning demonstrates the widest spread of CGPA values, suggesting greater variability in academic outcomes. In contrast, Hybrid learning shows a more compact distribution with tighter clustering around the median, indicating more consistent academic performance among students who adopt this learning mode.
- **The stacked bar chart** highlights a clear trend in learning mode preference across years of study, showing increased adoption of Hybrid learning among Year 3 and Year 4 students. This trend suggests that as students advance academically, they increasingly favour flexible learning arrangements that can accommodate heavier workloads, complex coursework, and time management demands.
- **The heatmap** illustrates the combined influence of learning mode and year of study on CGPA, revealing notable performance differences across student groups. An observable anomaly appears among Year 4 students who rely fully on Online learning, as they record the lowest average CGPA, while Year 1 Offline students demonstrate relatively strong academic performance. This suggests that fully online learning may pose greater challenges for senior students who face more advanced academic requirements.
- Overall, these findings connect to the original problem context by demonstrating that learning mode preferences, academic progression, and performance consistency are closely interrelated. While students are generally adaptable across learning environments, hybrid learning emerges as a preferred and stable option, particularly for senior students. The results also highlight the need for additional academic support for fully online senior students, reinforcing the importance of flexible yet well-supported learning strategies in enhancing student success at UMK.

5.4 Skills and Activities (Syazwanie)

- **The density plot** shows that both active and non-active students share a similar peak CGPA between 3.3 and 3.4, though active students exhibit a higher density in the elite (3.8 - 4.0) range. This trend suggests that campus engagement does not detract from studies but rather correlates with higher academic achievement. This finding supports the case study context that "active" lifestyles are often paired with the discipline required for top-tier performance.
- **The grouped bar chart** indicates that students with "High" skill hours perform better when they also participate in co-curricular activities, whereas "Medium" skill students see a drop in GPA when activities are added. This suggests that while top-tier students thrive on high engagement, moderate-engagement students may experience "over-commitment." This reinforces the case study problem that the impact of extra-curriculars depends heavily on a student's baseline capacity.
- **The stacked bar chart** reveals that students in the highest CGPA bracket (3.7–4.0) are most represented in "Soft Skills" and "Office & Engineering Tools," while the "Hardware" category is uniquely composed of students in the lowest CGPA range. This pattern highlights a distinct correlation between academic tiering and the type of skills students prioritize. This supports the problem context regarding how academic standing influences a student's professional development focus.
- **The line graph** shows a significant decline in CGPA for students with "High" skill development hours as they progress from Year 1 to Year 4, while "Low" hour students see a steady increase. This anomalous drop suggests that high-intensity skill-building may lead to academic burnout or time-management conflicts in later years. This finding identifies a critical trend regarding the long-term sustainability of high-pressure student lifestyles.
- **The correlation heatmap** identifies the "High Skill/Yes Co-curricular" intersection as the peak performance zone (3.57) and the "Medium Skill/Yes Co-curricular" as the lowest (3.37). This numerical breakdown confirms that synergy between skills and activities is only beneficial when managed at a high level of efficiency. This analysis supports the study context that academic success is a product of how effectively different lifestyle factors are integrated.
- In conclusion, the results show that, while high involvement in skills and activities corresponds with elite academic performance, it also increases the risk of burnout over time. The findings imply that academic success is not only about involvement, but the strategic balance of commitments. For many students, over-extending into too many areas might lead to a decrease in stability and performance.

6.0 INTERACTIVE VISUALIZATION DEPLOYMENT

6.1 Description of the Streamlit Application Architecture

- **Main Entry Point (main.py)**: Acts as the core controller of the application. It handles page configuration, navigation and routing between different analytical objectives.
- **Home Page (home.py)**: Serves as the landing page introducing EduTrack, its objectives. It provides users with a real-time data overview before navigating to analytical sections and embedded with Google Form Questionnaire.
- **Objective Modules (objective1_*.py, objective2_*.py, etc.)**: Each objective file represents a dedicated analytical module, focusing on specific aspects such as demographic characteristics, academic factors, etc.
- **Dataset Layer (cleaned_student_performance.csv)**: The application uses a preprocessed and cleaned dataset to ensure efficient loading and consistent analysis. Data is loaded using Pandas and shared across visualizations.
- **Visualization Layer**: Interactive charts are generated using libraries such as Plotly, Plotly.express, Numpy, which integrate seamlessly with Streamlit to provide responsive and dynamic visual outputs.
- **Deployment Platform**: The application is deployed on Streamlit Community Cloud, allowing user access through a web browser without requiring local installation.

6.2 Interactive Features Implemented

- **Dropdown Filters (Selectbox / Multiselect)**: Users can filter data by demographic variables such as gender, faculty, or living arrangement.
- **Sliders**: Sliders are used to adjust numeric ranges such as age, allowing users to observe performance trends dynamically across different ranges of ages.
- **Checkbox Controls**: Checkboxes allow users to toggle additional insights such as summary statistics, interpretations, or distribution descriptions without cluttering the interface.
- **Interactive Charts with Tooltips**: Visualizations generated using Plotly include hover tooltips that display precise values (e.g., CGPA, frequency, percentages), improving data interpretability.
- **Real-Time Updates for Homepage**: All visualizations in the homepage only update instantly when user inputs change, supporting exploratory analysis and immediate feedback.
- **Embedded Google Form in Homepage**: Users can answer using the Streamlit page and immediately get the real-time update visualization in home.py.

7.0 CONCLUSION AND FUTURE WORK

The EduTrack project successfully implemented a data-driven framework for analysing student academic performance at Universiti Malaysia Kelantan (UMK) using scientific visualisation. The data reveal that academic achievement is a multidimensional outcome driven by the synergy of disciplined study habits, consistent attendance, and excellent time management, rather than just socioeconomic position. . Notably, the data revealed that students from B40 and M40 households have remarkable academic resilience, frequently exceeding their higher-income counterparts. While active participation in co-curricular activities and skill development is associated with top performance, the study also discovered a "burnout threshold," where over-commitment in later years of study might result in a drop in CGPA. Furthermore, the high preference for hybrid learning suggests a clear need for educational flexibility that strikes a compromise between convenience and performance stability. Finally, this experiment demonstrates that interactive visualisation is a critical tool for converting complex survey data into usable insights, allowing educators to build focused interventions that promote student achievement.

Building on these findings, future study should aim to broaden the EduTrack platform's research scope and technical capabilities. Increasing the sample size to encompass a more diverse representation of all UMK campuses will increase the results' generalisability. There is also a tremendous possibility to move from descriptive to predictive analytics by using machine learning models capable of detecting "at-risk" students early in the semester. A longitudinal strategy, which follows specific student cohorts from their first year to graduation, would provide a better understanding of how academic pressure and lifestyle changes evolve over time. Furthermore, the platform might benefit from real-time integration with university administration systems and the creation of automated feedback tools to assist students in maintaining a healthy academic-life balance. By expanding this research to additional Malaysian universities, the findings could become a national benchmark for improving student outcomes across the higher education sector.

8.0 REFERENCES

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